Resource-Aware Edge AI for Malaria Cell Detection: Precision-Resolution Tradeoffs and Tiered TensorRT Inference

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Abstract — Computer-assisted malaria microscopy must preserve infected-cell sensitivity while operating under limits on computation, latency, and energy. This paper presents a biomedical image-analysis case study that jointly evaluates numerical precision and input resolution for edge deployment. A YOLOv8s detector was trained on synthetic 1280 × 720 microscopy pages generated from public infected and uninfected red-blood-cell cutouts. Eight TensorRT configurations—FP16 and INT8 at 320, 480, 640, and 960 pixels—were evaluated on 200 synthetic test pages containing 7,469 labeled cells, without source-cell-level separation between training and testing. FP16-640 achieved the highest infected-cell recall (97.5%) at 17.5 ms and 0.282 J per inference. FP16-480 retained 96.8% recall while reducing latency to 11.4 ms and energy to 0.243 J, and INT8-320 minimized latency and energy at 7.6 ms and 0.198 J with 94.3% recall. These measured operating points were integrated into a Jetson Orin Nano Super prototype with confidence- and battery-gated escalation, infected-cell fraction reporting, suspicious-cell magnification, selective Eigen-CAM saliency, and compact LCD output. The nonmonotonic precision-resolution results show that deployment settings should be measured jointly on the target device. This work is a systems proof of concept for resource-constrained biomedical image screening, not a clinical validation.

Index Terms — *biomedical image analysis, computer-aided detection, edge AI, malaria microscopy, TensorRT, quantization, energy efficiency*

I. INTRODUCTION

Malaria microscopy supports parasite detection, species and stage identification, and estimation of parasite density in stained blood films [1]. Reliable interpretation, however, depends on trained personnel, consistent slide preparation, suitable optics, and quality assurance. These requirements motivate computer-assisted tools that support rather than replace human review.

Prior deep-learning studies have demonstrated classification and localization of infected red blood cells [2]–[4], but many reports emphasize predictive metrics obtained on desktop hardware. A deployable biomedical system must also account for power availability, processing delay, model footprint, camera compatibility, and the communication of uncertain predictions. Numerical precision and input resolution are especially important because they jointly affect small-object visibility, accelerator utilization, memory traffic, and energy consumption.

This study evaluates the combined effect of TensorRT precision and input resolution for malaria-cell detection on an NVIDIA Jetson Orin Nano Super. The contribution is an end-to-end biomedical computing systems study rather than a new detector architecture. Public single-cell images are converted into dense detection pages; a YOLOv8s model is trained and exported to eight FP16/INT8 engines; infected-cell performance, latency, and whole-device energy are measured; and three operating points are integrated into a camera-to-LCD prototype.

The paper contributes: (1) a reproducible synthetic microscopy-page pipeline with automatic object labels; (2) a benchmark of FP16 and INT8 at four resolutions; (3) evidence-based selection of high-sensitivity, balanced, and low-energy operating points; and (4) a human-in-the-loop interface with tier status, infected-cell fraction, suspicious-cell zoom, and selectively triggered Eigen-CAM. The reusable contribution is the deployment workflow; adaptation to another biomedical task would require new data, retraining, and independent validation.

II. RELATED WORK

Rajaraman et al. released a widely used set of segmented parasitized and uninfected cell images and evaluated transfer-learning classifiers [2]. Poostchi et al. reviewed malaria image analysis and emphasized persistent challenges involving segmentation, staining variability, acquisition conditions, and generalization [3]. Hung et al. applied Faster R-CNN to malaria fields of view, showing the value of localization and counting beyond image-level classification [4]. The present work addresses a complementary question: how precision and resolution affect embedded deployment, and how measured operating points can be translated into an integrated biomedical edge prototype.

TensorRT supports optimized FP16 and calibrated INT8 inference on NVIDIA hardware [5]. Reduced precision can improve throughput and energy efficiency, but the accuracy cost depends on model architecture, calibration data, input size, and hardware. Precision and resolution should therefore be evaluated together on the target device. Eigen-CAM extracts a saliency map from principal components of intermediate activations without gradients or class-relevance scores [6]. Because it is class agnostic, this study displays it only within boxes already predicted as infected.

III. MATERIALS AND METHODS

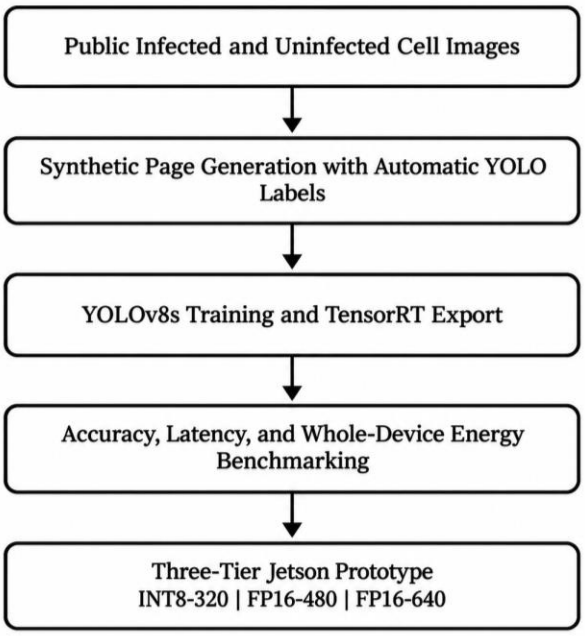


Fig. 1. Experimental workflow from public cell images through TensorRT benchmarking and three-tier edge deployment.

A. Biomedical Image Data and Synthetic Pages

The source data were National Library of Medicine/NIH thin-blood-smear cell images introduced by Rajaraman et al. [2]. Infected and uninfected images were converted to RGBA cutouts using background removal and retained in separate class directories. A quality-control tool converted normalized YOLO labels back to pixel coordinates and rendered sampled previews so automatically generated annotations could be checked visually.

The generator created 2,000 training pages, 200 validation pages, and 200 test pages at 1280 × 720 pixels. Each page contained 25–55 cells with approximately balanced class sampling. Cutouts were resized, rotated, and placed with maximum pairwise IoU 0.03 to limit unrealistic overlap. A fixed seed of 42 supported reproducibility. Texture, vignette, brightness/contrast and color changes, blur, and JPEG artifacts increased visual diversity. The test set contained 7,469 labeled instances: 3,824 uninfected and 3,645 infected.

The composite pages were independently generated, but source-cell images were not partitioned before synthesis. Augmented instances of a source cell may therefore occur across training and test pages. This overlap can inflate absolute performance relative to unseen patients or slides; the benchmark is used primarily for controlled comparison among deployment configurations.

B. Detector Training and TensorRT Export

A COCO-pretrained YOLOv8s detector was fine-tuned at 640-pixel input size using Ultralytics [7]. Training used automatic batch selection, automatic mixed precision, deterministic seed 0, and standard augmentation. The run was configured for 300 epochs with early-stopping patience 50, terminated after 212 epochs, and retained a best checkpoint near epoch 162. Training required approximately 13 h 22 min. The best validation aggregate reached 97.99% precision, 97.99% recall, 99.25% mAP50, and 98.24% mAP50–95.

The checkpoint was exported to TensorRT at 320, 480, 640, and 960 pixels in FP16 and post-training INT8 forms. INT8 calibration used training images rather than the test split. All eight engines were evaluated on the same 200-page test set. Infected-class precision and recall were emphasized because missed infected cells are a sensitivity concern; mAP50–95 was also recorded.

C. Edge Platform and Energy Measurement

Experiments ran on an NVIDIA Jetson Orin Nano Super Developer Kit with JetPack 6.2.1 [8]. Accuracy and TensorRT inference latency were measured using Ultralytics validation of each engine; reported latency does not represent complete camera-to-display pipeline time. Energy was measured separately using the same prerecorded video across configurations. The LCD was disabled. TegraStats logged the VDD_IN rail, and current power samples were integrated over timestamp intervals, $E = \Sigma(P \cdot \Delta t)$, then divided by completed inferences. The values therefore represent whole-device energy per inference, not isolated GPU energy. Each energy configuration was measured once.

Independent recomputation from raw logs matched recorded per-inference energies to within approximately 0.3%, supporting arithmetic consistency. This verification does not replace repeated trials or fixed clock-state control, but it reduces the risk that the reported ordering resulted from a parsing or unit-conversion error.

D. Three-Tier Prototype and Review Interface

Three measured endpoints were deployed concurrently: Tier 3 INT8-320, Tier 2 FP16-480, and Tier 1 FP16-640. For each frame, a confidence score combines mean infected-detection confidence (70%) with the mean of the five highest detection confidences (30%) and is exponentially smoothed. The controller selects an operating tier based on this score and the available battery state: scores below 0.90 can trigger Tier 2 when battery state is above 20%, while scores below 0.75 can trigger Tier 1 when battery state is above 60%. As the smoothed confidence changes, the active tier can change accordingly. The demonstration used a mocked battery value, so the controller is prototype integration rather than a validated energy-saving policy.

The interface reports infected and uninfected counts, an estimated infected-cell fraction (parasitemia proxy), active tier, and smoothed confidence on a 240 × 320 LCD. It magnifies the lowest-confidence infected detection and tracks the target briefly to reduce visual jumping. A standard USB webcam provided prototype input; a USB microscope could replace it only after field-of-view, illumination, scale, and accuracy validation. Selective Eigen-CAM uses a separate PyTorch checkpoint because TensorRT engines do not expose intermediate activations. The map is masked to predicted infected boxes and is

interpreted as a qualitative feature-response aid, not a causal medical explanation.

IV. RESULTS

A. Precision-Resolution Benchmark

TABLE I.

TENSORRT TEST PERFORMANCE AND DEPLOYMENT COST

P_{inf} and R_{inf} denote infected-class precision and recall.

Config	P_{inf}	R_{inf}	mAP50-95	Latency (ms)	Energy (J/inf.)
INT8-320	0.954	0.943	0.910	7.6	0.198
FP16-320	0.961	0.952	0.940	9.0	0.225
INT8-480	0.974	0.954	0.956	10.4	0.226
FP16-480	0.976	0.968	0.974	11.4	0.243
INT8-640	0.975	0.969	0.964	10.6	0.256
FP16-640	0.984	0.975	0.976	17.5	0.282
INT8-960	0.985	0.951	0.969	16.2	0.344
FP16-960	0.986	0.974	0.978	21.4	0.464

FP16-640 achieved the highest infected-cell recall at 97.5%, with 98.4% precision, 17.5 ms latency, and 0.282 J per inference. FP16-480 retained 96.8% recall while lowering latency by 34.9% and energy by 13.8% relative to FP16-640. INT8-320 minimized cost at 7.6 ms and 0.198 J, reducing latency by 56.6% and energy by 29.8% relative to FP16-640, with a 3.2-percentage-point recall reduction.

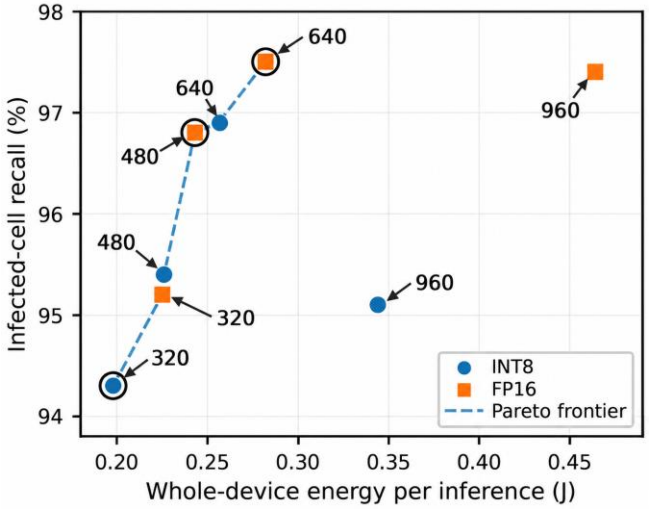


Fig. 2. Whole-device energy versus infected-cell recall. The dashed line connects nondominated configurations; rings mark the selected tiers.

Performance was not monotonic with resolution. FP16-640 slightly exceeded FP16-960 in infected-cell recall while using 39.2% less energy. FP16-480 achieved lower energy than INT8-640 at nearly identical recall. The 960-pixel configurations offered no recall advantage sufficient to justify their additional energy cost. Small differences in infected-cell recall should be interpreted cautiously because confidence intervals were not estimated and source-cell images were not separated across training and test sets.

B. Operating-Point Selection and Prototype

The benchmark supports three deployment roles. FP16-640 is the high-sensitivity endpoint. FP16-480 is the balanced endpoint because it preserves nearly all infected recall while substantially lowering latency and energy. INT8-320 is the low-energy endpoint. The selected tiers represent separated operating roles rather than simply the three highest-recall configurations.

FP16-480 is particularly important: it loses only 0.7 percentage points of recall relative to FP16-640 while saving 6.1 ms and 0.039 J per inference. INT8-640 was not selected as the balanced tier because it was only 0.8 ms faster and 0.1 percentage point higher in recall than FP16-480, yet consumed 5.3% more energy. The benchmark shows how a target-device measurement campaign can reduce several plausible model variants to a small, interpretable set of runtime choices.

At the opposite extreme, FP16-960 consumed 64.5% more energy than FP16-640 without improving infected-cell recall. The 960-pixel configurations were therefore dominated by smaller operating points for this device and benchmark. Separate FP32 energy-only baselines measured 0.522 J at 640 pixels and 0.685 J at 960 pixels; INT8-320 used 62.1% and 71.1% less energy than those baselines, respectively, although FP32 accuracy is not reported because validation runs were unstable on the device.

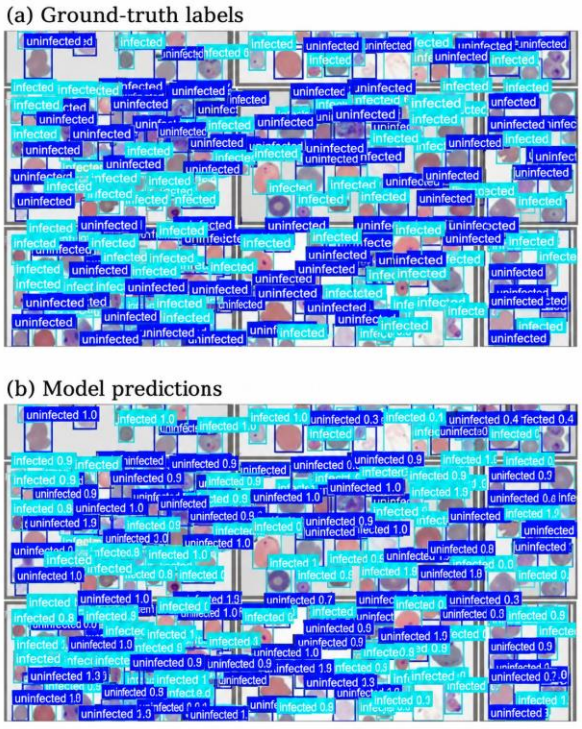


Fig. 3. Representative synthetic montage: generated ground-truth boxes (left) and retained-model predictions (right).

Across dense synthetic scenes, predicted boxes visually followed the generated annotations, including small and closely spaced cells. The visualization illustrates why object detection, rather than page-level classification, is required

for counting infected and uninfected cells and estimating infected-cell fraction.

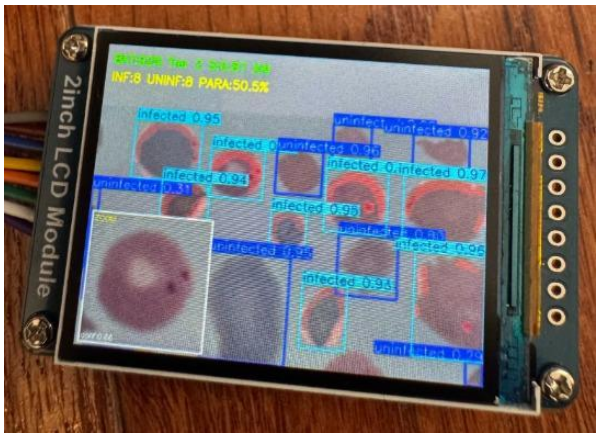


Fig. 4. Physical prototype showing detections, parasitemia proxy, tier status, suspicious-cell magnification, and masked Eigen-CAM saliency.

In a 43.70-s demonstration, the complete application read 264 frames (6.04 end-to-end frames/s) and executed 590 model invocations (13.50 invocations/s), showing that many frames progressed through multiple tiers. Eigen-CAM recomputed on 11 of 264 frames (4.2%). PyTorch forward plus SVD averaged 272.11 ms per trigger and amortized to 11.34 ms per processed frame. This is runtime accounting, not a matched no-XAI slowdown experiment.

The physical demonstration establishes software and hardware integration rather than diagnostic performance. It confirms that detection, tier switching, count aggregation, estimated infected-cell fraction, suspicious-cell magnification, saliency generation, and LCD rendering can operate in one application on the target device. Because the input was a webcam demonstration rather than labeled microscope video, the observed frame rate and tier transitions should not be interpreted as clinical throughput or error-recovery measurements.

V. DISCUSSION

The principal finding is that numerical precision and input resolution must be co-selected on the target device. Higher resolution did not uniformly improve infected-cell recall, and INT8 did not dominate FP16 at every input size. FP16-480 approached the sensitivity of FP16-640 while reducing both latency and whole-device energy. These results support evaluating multiple operating points rather than deploying the largest or lowest-precision model by default.

The prototype translates the benchmark into a human-in-the-loop biomedical design. Starting with a low-cost tier and escalating uncertain frames provides a framework for balancing resources with review quality. However, the cascade has not been compared with fixed-tier baselines on labeled video; the paper therefore does not claim measured cascade energy savings or error recovery. The validated contribution is the implementation and evidence-based choice of its tiers.

The review interface is designed to expose uncertainty rather than hide it. Counts and estimated infected-cell fraction provide a compact summary, the zoom panel surfaces the lowest-confidence infected detection, and the saliency overlay shows strong feature response within infected boxes. These elements could help a technician prioritize review after proper microscopy validation. They do not make the prototype an autonomous diagnostic system.

A technician-facing system may benefit from distinct operating modes rather than one globally optimal model. The low-energy tier can support rapid scanning or constrained battery states, the balanced tier can handle routine review, and the high-sensitivity tier can be reserved for uncertain frames. The present cascade operationalizes this concept, while future labeled-video experiments are needed to determine whether the confidence and battery thresholds are effective.

A. Interpreting Whole-Device Energy

VDD_IN intentionally captures a broader quantity than accelerator-only power: CPU, GPU, memory, and other active board components contribute to the measured energy required to complete inference. This system-level measure is relevant to battery-oriented deployment. The LCD was disabled during these measurements, so the reported values represent compute-platform energy rather than complete product energy.

The ordering of measured configurations also shows why precision labels alone are insufficient. INT8-640 consumed more energy than FP16-480 while providing nearly identical infected-cell recall, and INT8-960 was dominated by several smaller configurations. Compiler behavior, memory traffic, preprocessing, and device utilization can outweigh the nominal arithmetic advantage of lower precision.

B. Transferability and Human-in-the-Loop Use

The prototype used a conventional USB video interface rather than a camera-specific software stack. A USB digital microscope or a camera coupled to an optical microscope could replace the demonstration webcam without changing the detector, TensorRT benchmark harness, tier controller, energy parser, or LCD interface. Scientifically, however, that transition would require validation under controlled magnification, focus, illumination, staining, and cell scale.

For another image-screening task, a developer could reuse the sequence of data preparation, multi-resolution FP16/INT8 export, common-test-set evaluation, VDD_IN integration, Pareto-style operating-point selection, uncertainty display, and conditional saliency. The trained weights, labels, threshold logic, and output interpretation would still require task-specific replacement and validation. Transferability here means reuse of the engineering workflow, not cross-disease diagnostic generalization.

VI. THREATS TO VALIDITY

Construct validity is limited because infected-cell recall on synthetic pages is not equivalent to patient-level diagnostic sensitivity. Likewise, the estimated infected-cell fraction is the fraction of detected cells classified as infected

in the current field, not a standardized clinical parasite-density measurement. VDD_IN captures whole-device energy but does not isolate individual subsystems.

Internal validity is supported by evaluating all eight engines on the same test pages and using prerecorded input for energy comparison. It is weakened by one energy trial per configuration, unrecorded historical power and clock settings, and possible differences in saved validation command defaults. Independent raw-log recomputation supports numerical consistency but cannot remove these experimental uncertainties.

External validity is limited by source-instance overlap and the absence of independent microscope fields across patients, stains, laboratories, and imaging devices. The physical prototype demonstrates software and hardware integration, not clinical generalization. Conclusion validity is also limited for the controller and XAI components because they were demonstrated without matched fixed-tier or no-XAI ablations.

VII. LIMITATIONS AND FUTURE WORK

The largest limitation is source-instance overlap across synthetic splits. Independently generated pages may contain augmented versions of the same underlying source cells, so the reported 97.5% recall cannot be interpreted as performance on unseen patients, slides, laboratories, or microscopes. The current benchmark supports relative comparison among deployment configurations, not clinical sensitivity estimation.

Energy results were based on one run per configuration, and historical power-mode and clock settings were not recorded. Repeated trials should fix the power mode, exclude warm-up, report uncertainty, and include the display and imaging device when estimating full-product energy. The controller used mocked battery state and was not evaluated on labeled video. Future studies should quantify early termination, errors corrected by escalation, energy per processed frame, and sensitivity to thresholds.

The prerecorded input improved comparability among configurations but did not test optical acquisition variability. A USB microscope should be evaluated under controlled magnification, focus, illumination, staining, and slide preparation. Repeated measurements should also report confidence intervals and include the LCD and imaging device when estimating complete-product energy rather than compute-platform energy alone.

The system distinguishes infected and uninfected cells but does not identify Plasmodium species or developmental stage. Its parasitemia output is the fraction of detected cells classified as infected in the current field, not a validated clinical parasite-density protocol. The prototype is therefore not a diagnostic device and should not be used for autonomous treatment decisions.

A. Recommended Validation Sequence

A stronger second-stage study should separate four questions that are combined in the present proof of concept: biological generalization, imaging-domain transfer, reduced-precision behavior, and controller efficiency. First, source cells should be partitioned by patient or slide before any synthesis. Second, complete fields should be captured through a fixed microscope configuration. Third, the detector should be evaluated on an external site or device not represented during training. Fourth, repeated fixed-power measurements and labeled-video cascade experiments should report confidence intervals, early termination, escalation benefit, and energy per processed frame.

VIII. CONCLUSION

This work presents an end-to-end edge-AI case study connecting biomedical image synthesis, detector training, TensorRT quantization, whole-device energy measurement, operating-point selection, and physical deployment. Across eight configurations, FP16-640 delivered the highest infected-cell recall, FP16-480 provided a favorable balance of recall, latency, and energy, and INT8-320 minimized deployment cost. The nonmonotonic results show why precision and resolution should be benchmarked jointly on the target platform.

The selected engines were integrated into a functioning Jetson prototype with tiered inference, estimated infected-cell fraction (parasitemia proxy), suspicious-cell zoom, selective Eigen-CAM, and compact LCD output. Although clinical claims are limited by source-instance reuse, single-run power measurements, and the absence of independent microscope validation, the project establishes an end-to-end systems proof of concept and a reusable methodology for future resource-aware biomedical image-screening systems.

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